

Photovore

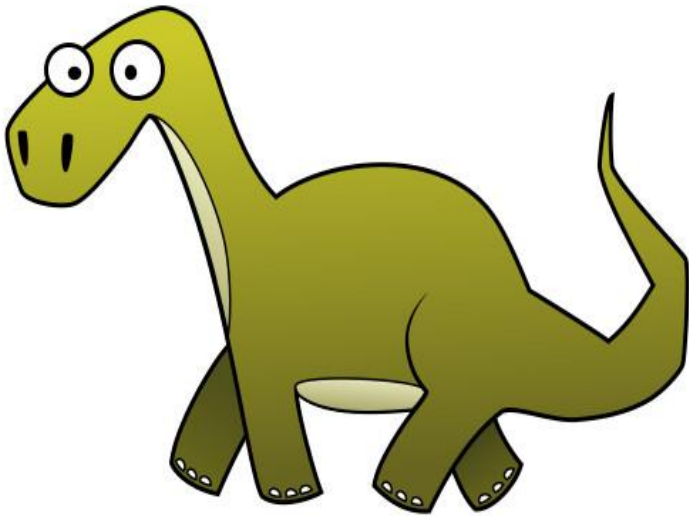
Fall 2026

Overview

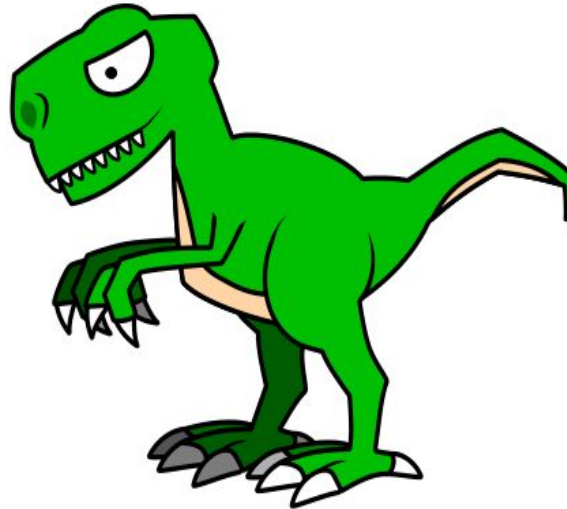
- Photovore Lab
- What is Electricity
 - Voltage
 - Current
- Basic Components
 - Wires
 - Resistors
- Photovore Circuit
 - Voltage Divider
 - Breadboards

The Lab

- Herbivore



- Carnivore

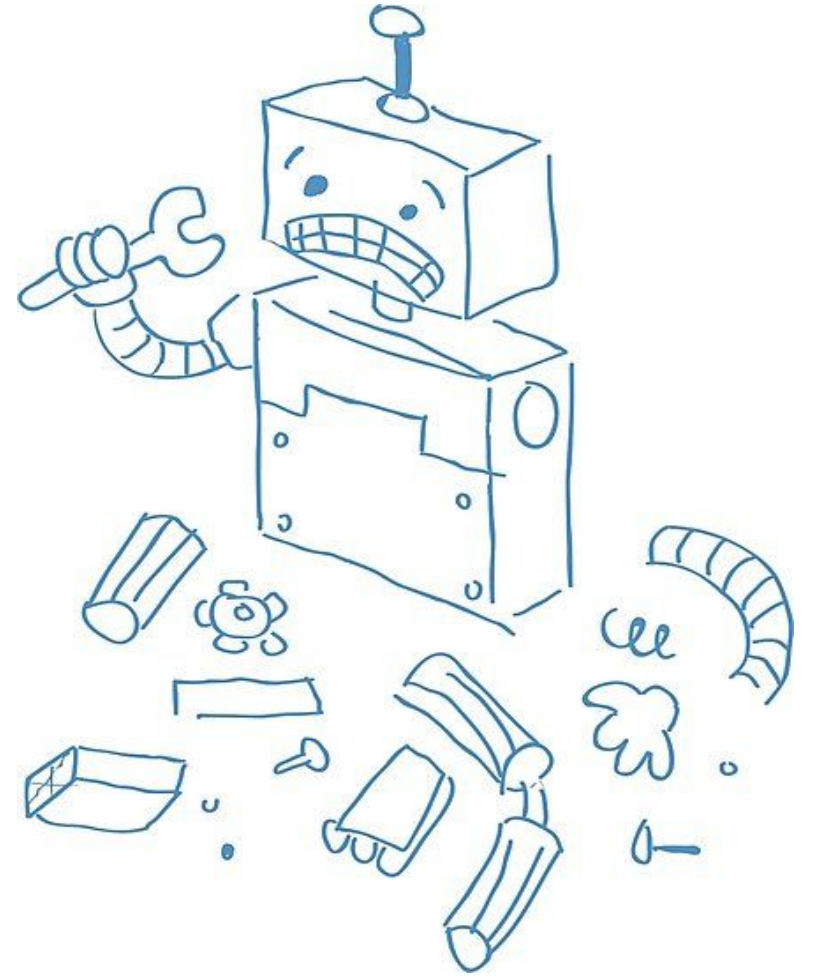


- Photovore?



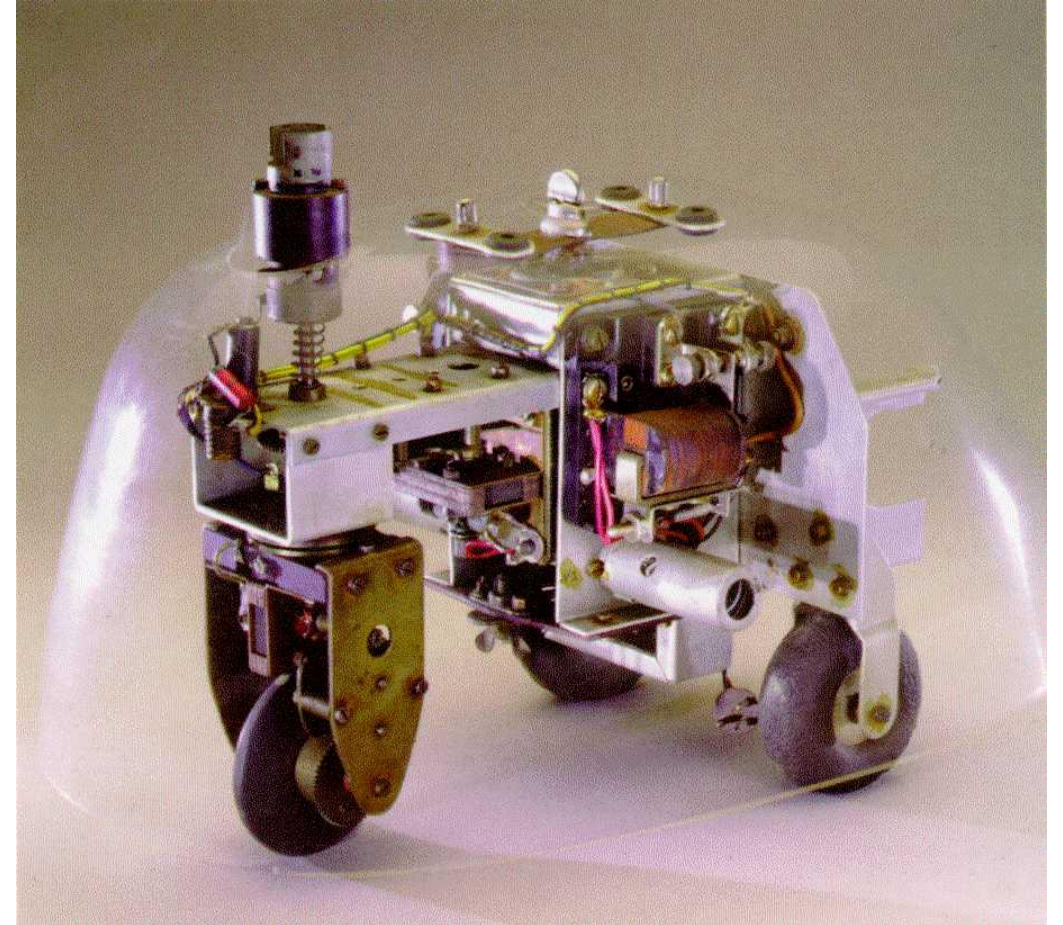
Lab Objectives

- Learn about sensors!
- Make a circuit with sensors!
- Make your robot sense light!
- Make your robot drive to the light!
- Make your robot eat the light!
 - but be careful...



History

- A photovore is a robot that is attracted to light.
- Grey Walter created the “turtle” robots in 1948 – they were among the first photovores.



History

Braitenberg published a paper (1984) on a simple method of controlling a photovore.



Braitenberg Vehicles

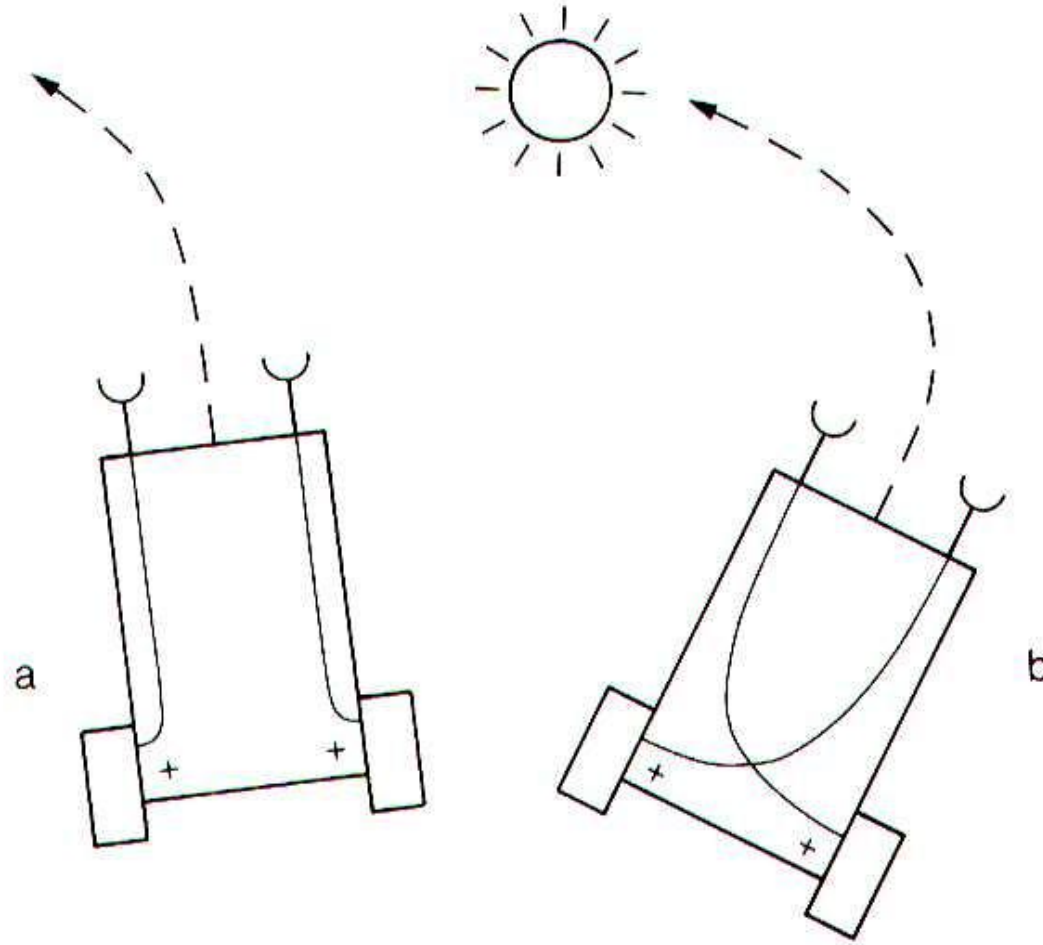


Figure 3

Vehicles 2a and 2b in the vicinity of a source (circle with rays emanating from it). Vehicle 2b orients toward the source, 2a away from it.

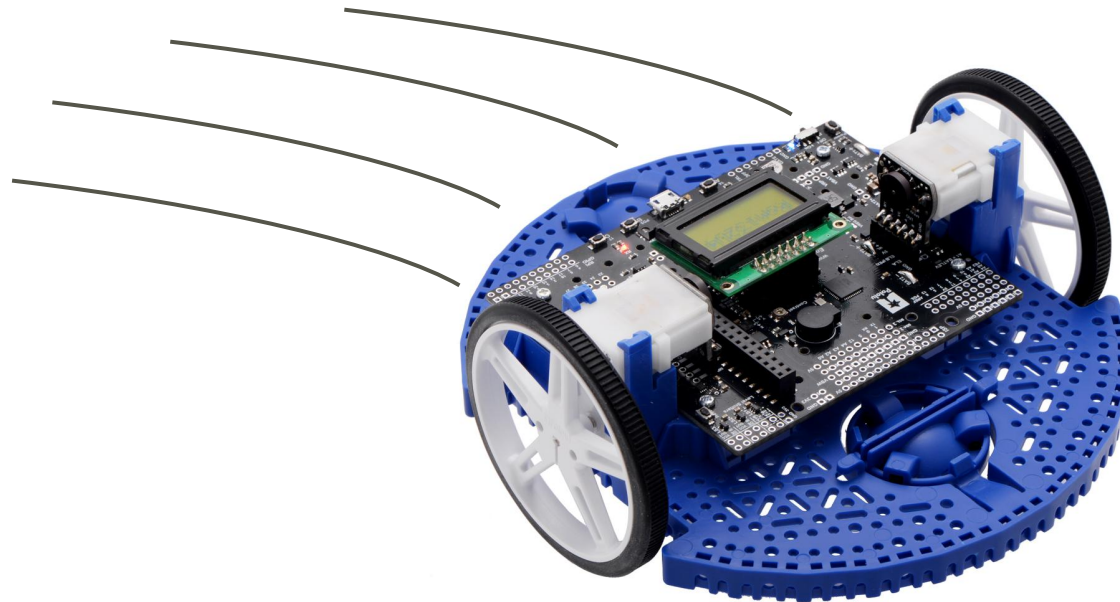
The Challenge

- Your task – to create a photovore.
- Your robot should be able to find the light as long as your robot is within 10(ish) feet of the light.
- After that, your sensors just can't tell the light from room light.
 - Or can they...
- Once you've found it, drive towards it!
- "Completeness" is being able to drive from any starting position to the light.
- You don't have to gracefully stop, try to though.
- You can't just wander around until you find it.

Circuits

Why Circuits

- The Romi's main chip can't measure light on its own
- We need to use circuits to convert light into something it can understand.



**WRONG
WAY**

What is Electricity

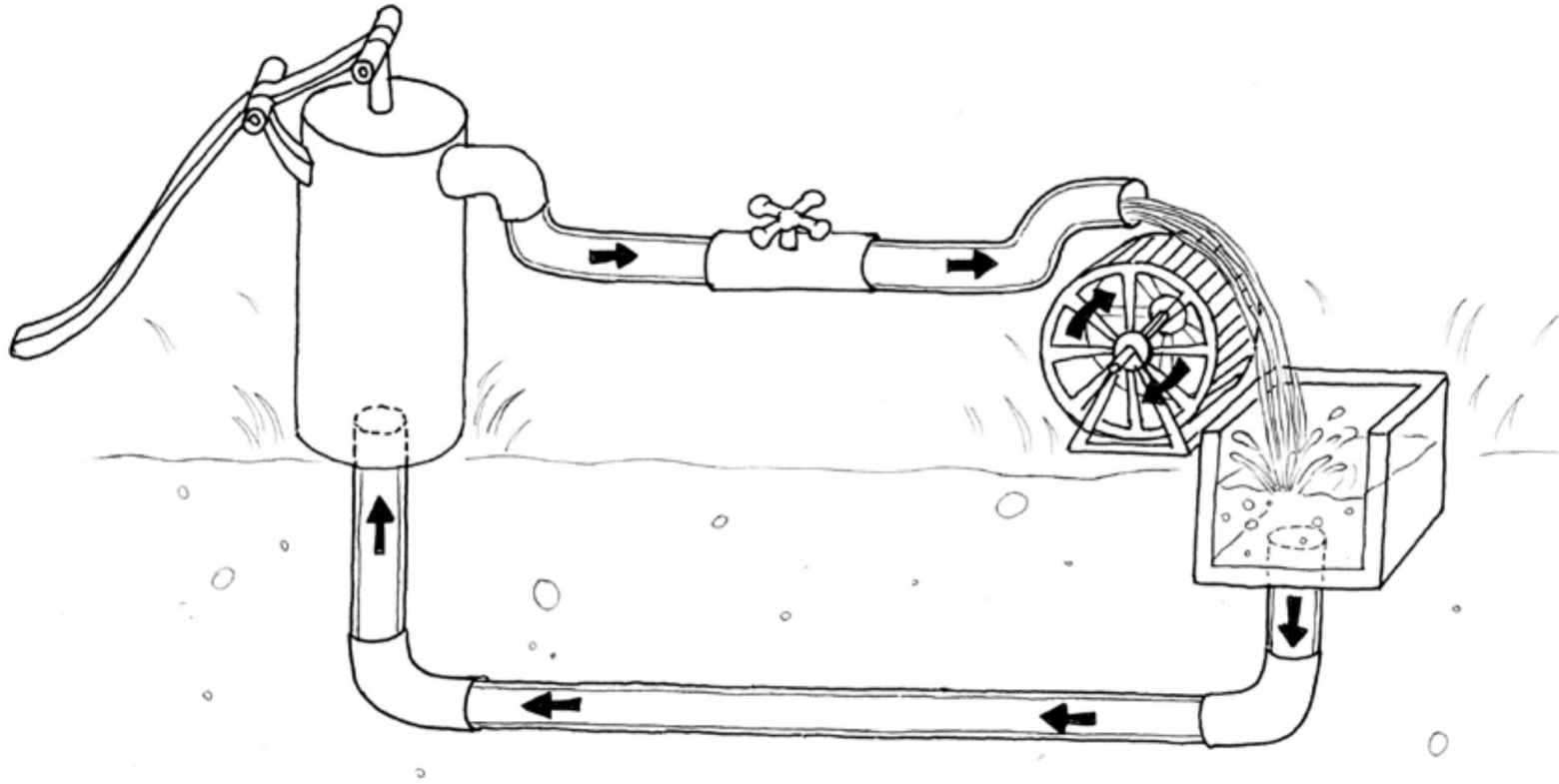
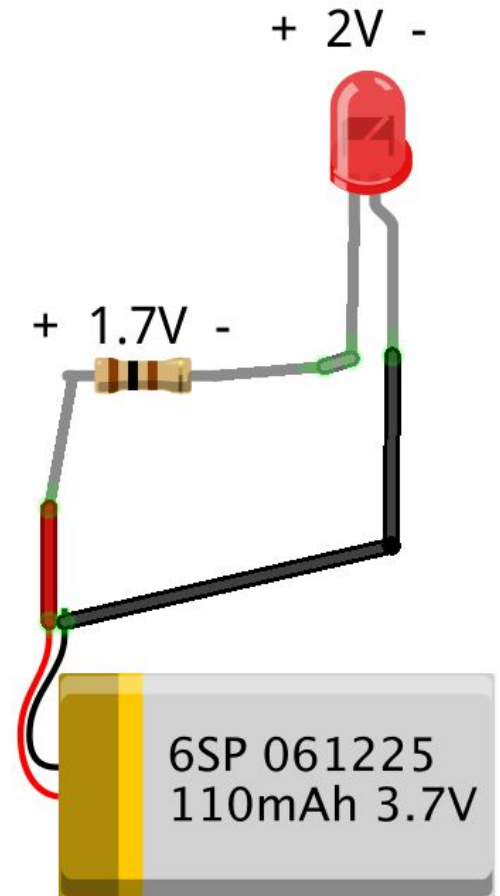
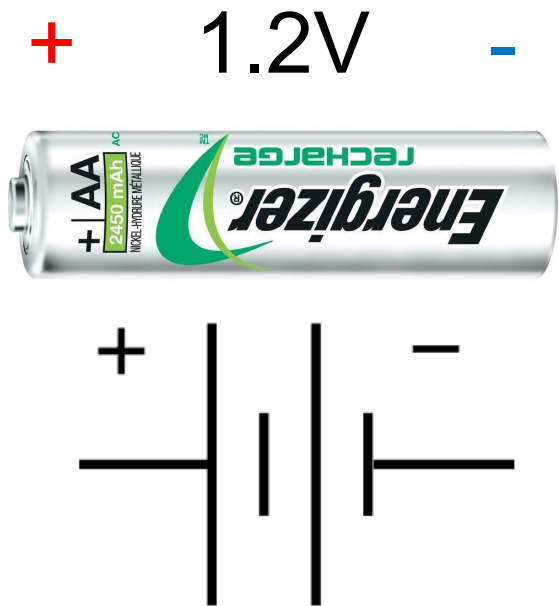


Image Credit: "Getting Started with Arduino" by Massimo Banzi

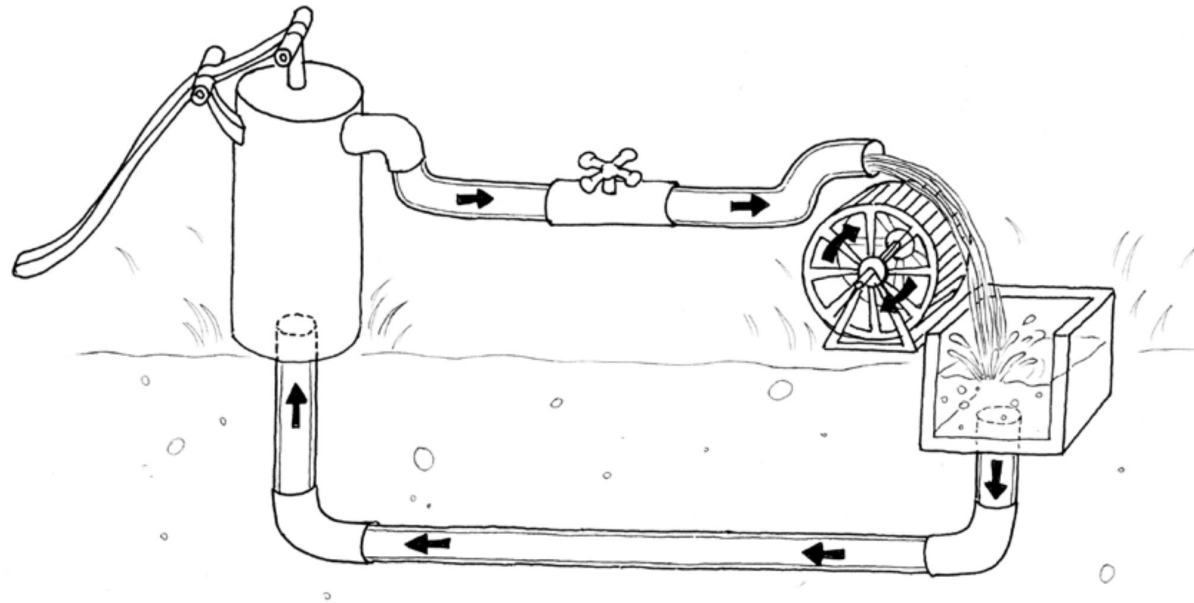
Voltage

- Voltage: Difference in electrical potential energy
 - Units: Volts



Candy Question!

- What is voltage in the water analogy?
- What are the units of voltage?
 - Bonus if you can break it down into basic SI units!



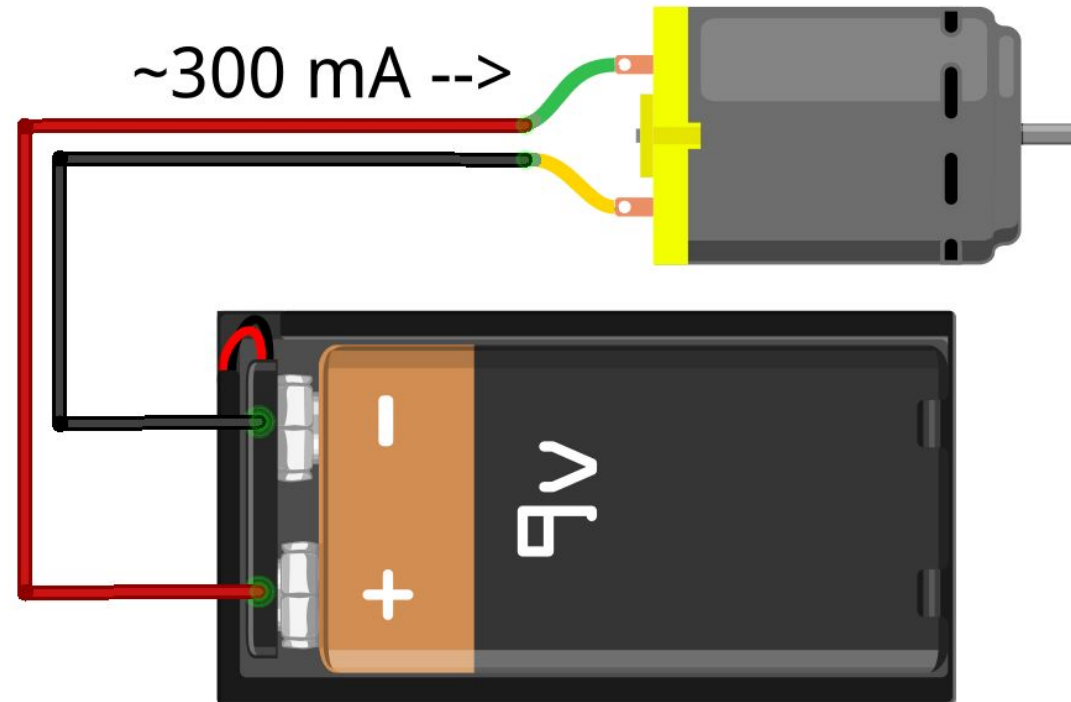
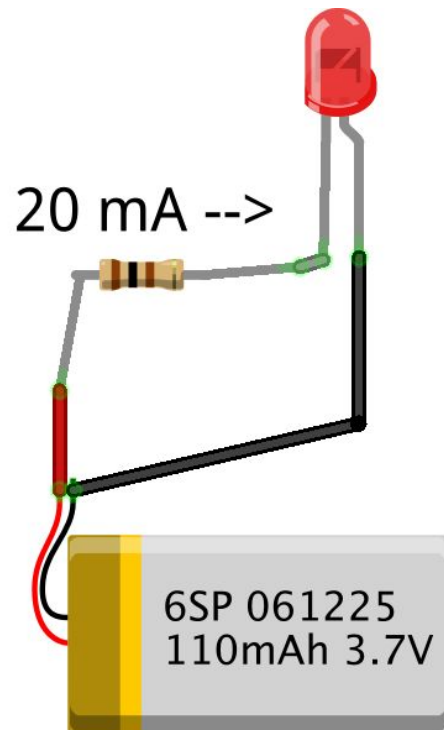
Answer:

- In the water analogy, the difference of potential energy of the water is the voltage.
 - For example, the decrease of energy as the water falls down from the tower, across the waterwheel, to a lower altitude would be like a decrease in voltage.
- Units: $\text{kg} * \text{meter}^2 / (\text{s}^3 \text{ A})$

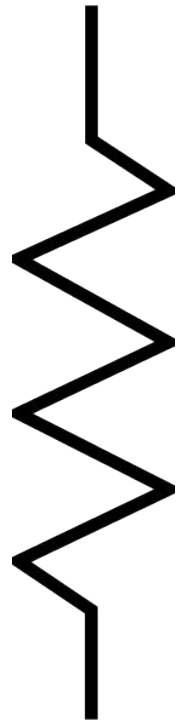


Current

- Current: rate of flow of electrical charge
 - Units: Amps (Coulomb/second) (1000 mA = 1 A)



Resistors

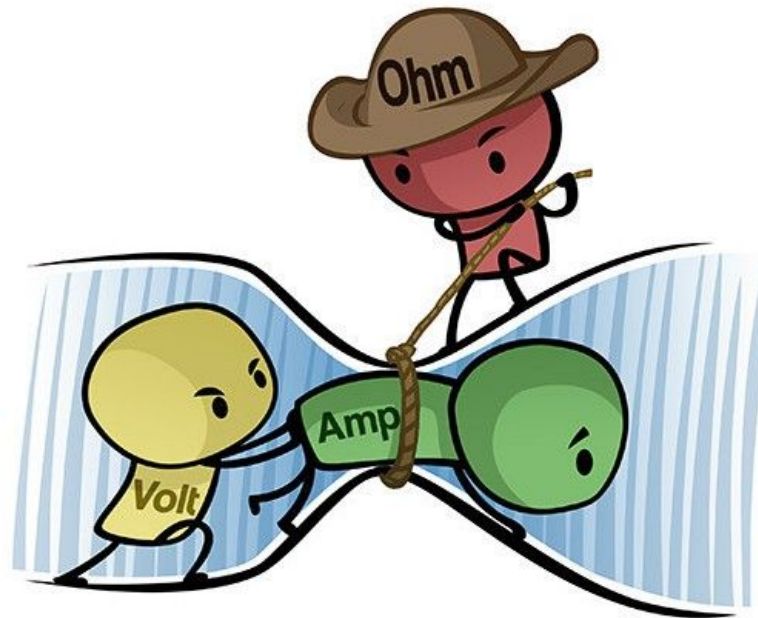


- Resist flow of electricity
- Colored bands describe the resistor's value
- Large values = Closer to an open circuit (air)
- Small values = Closer to a short circuit (wire)
- Units: Ohms

Review: Candy Question

Why does this cartoon make any sense at all?

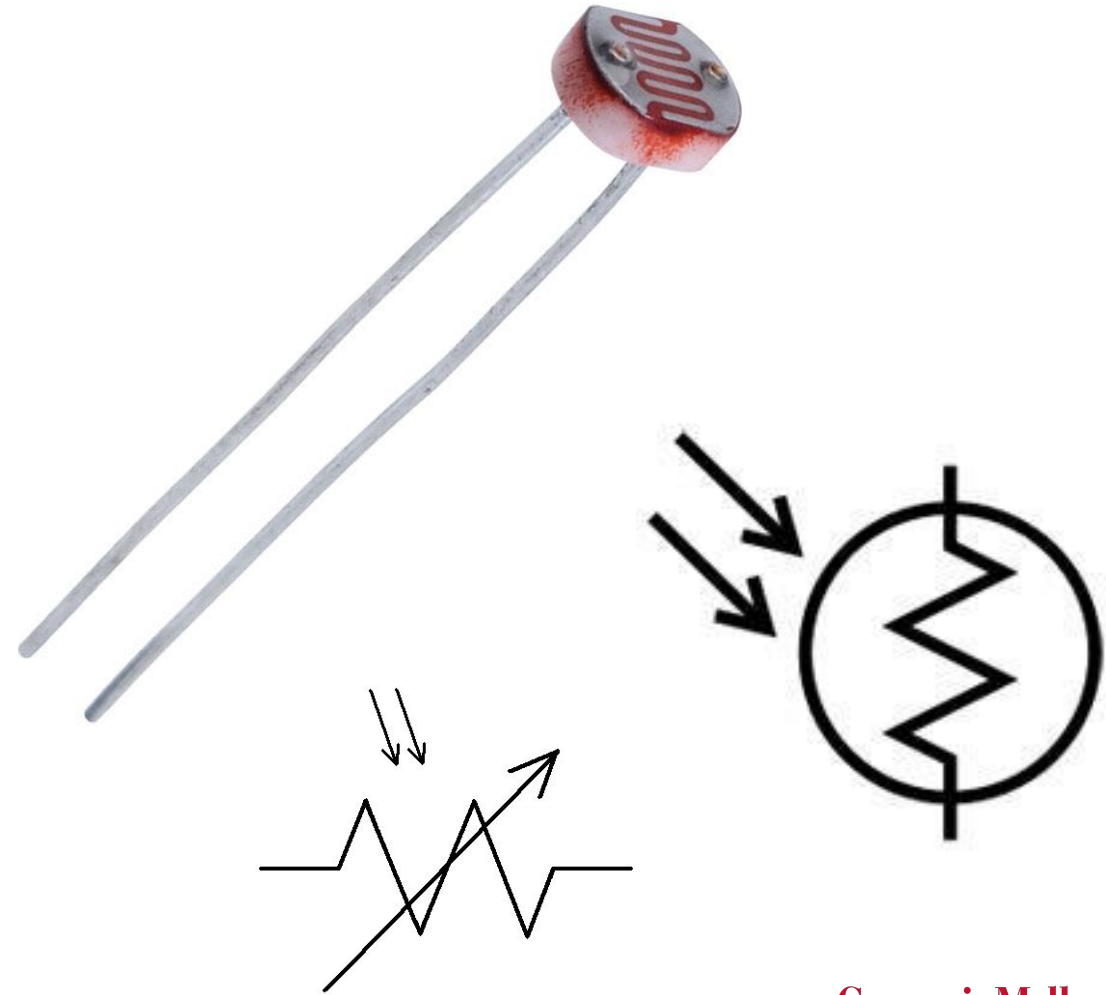
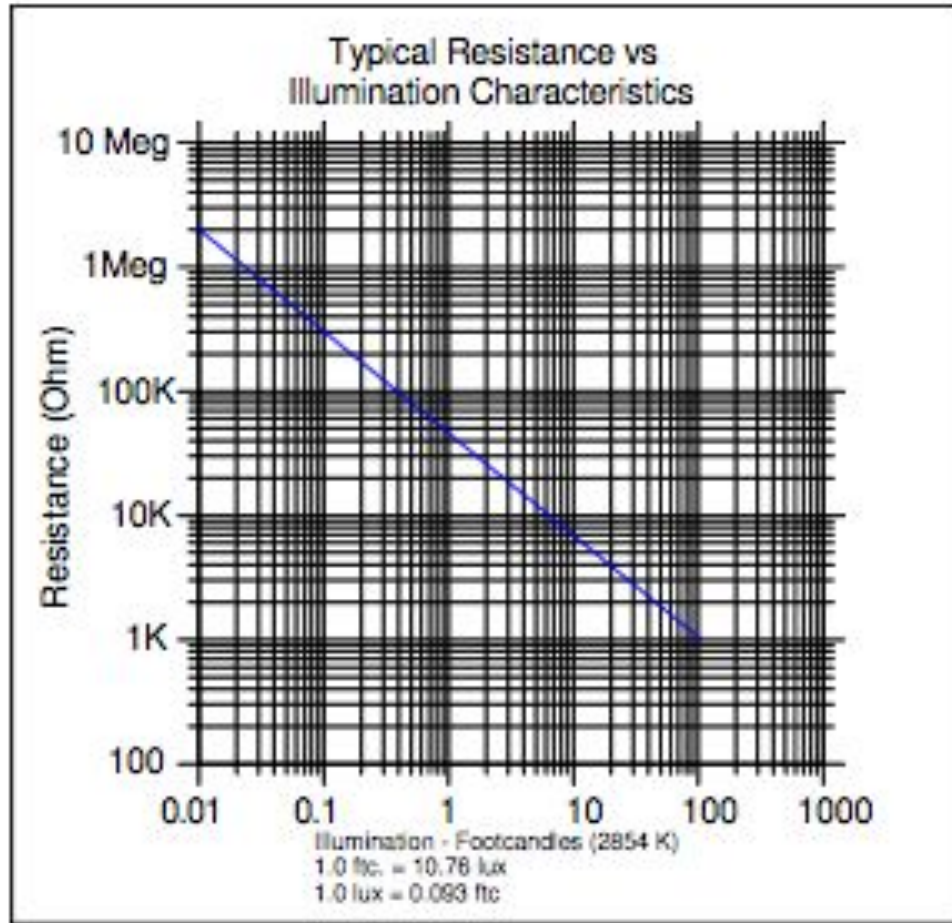
IE, how does this relate to voltage, current, and resistance?



Photoresistor

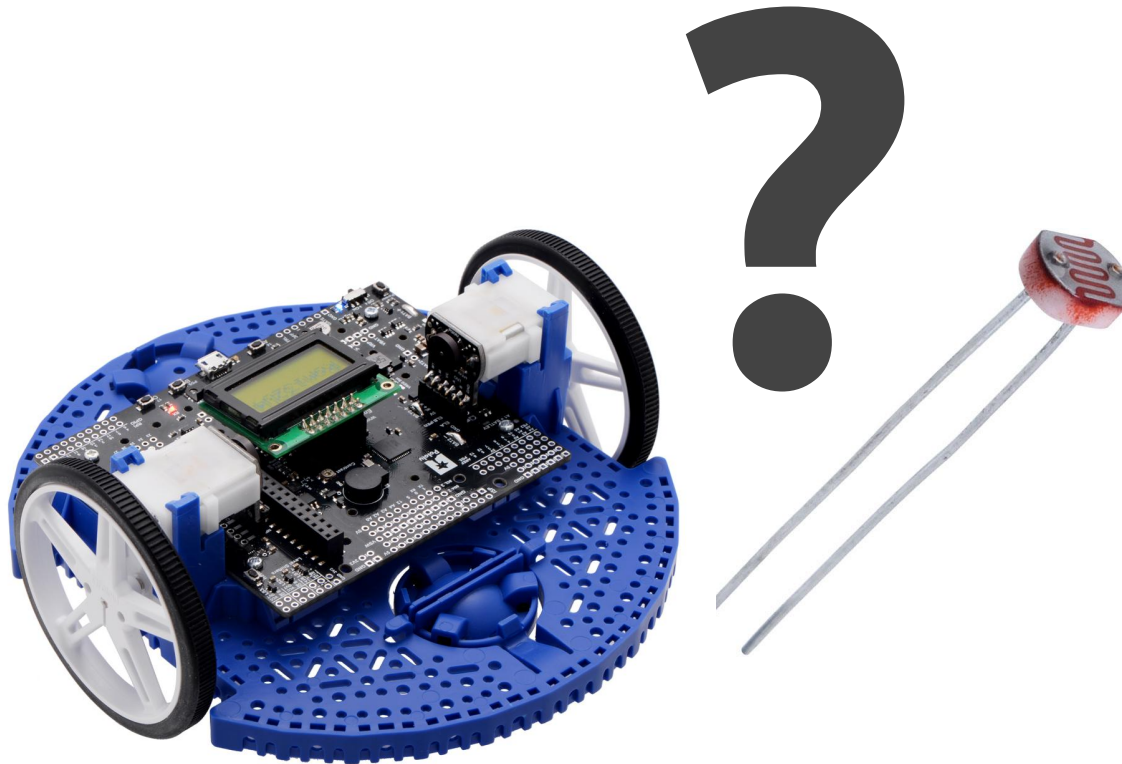
But none of this helps us measure light...

Photoresistor



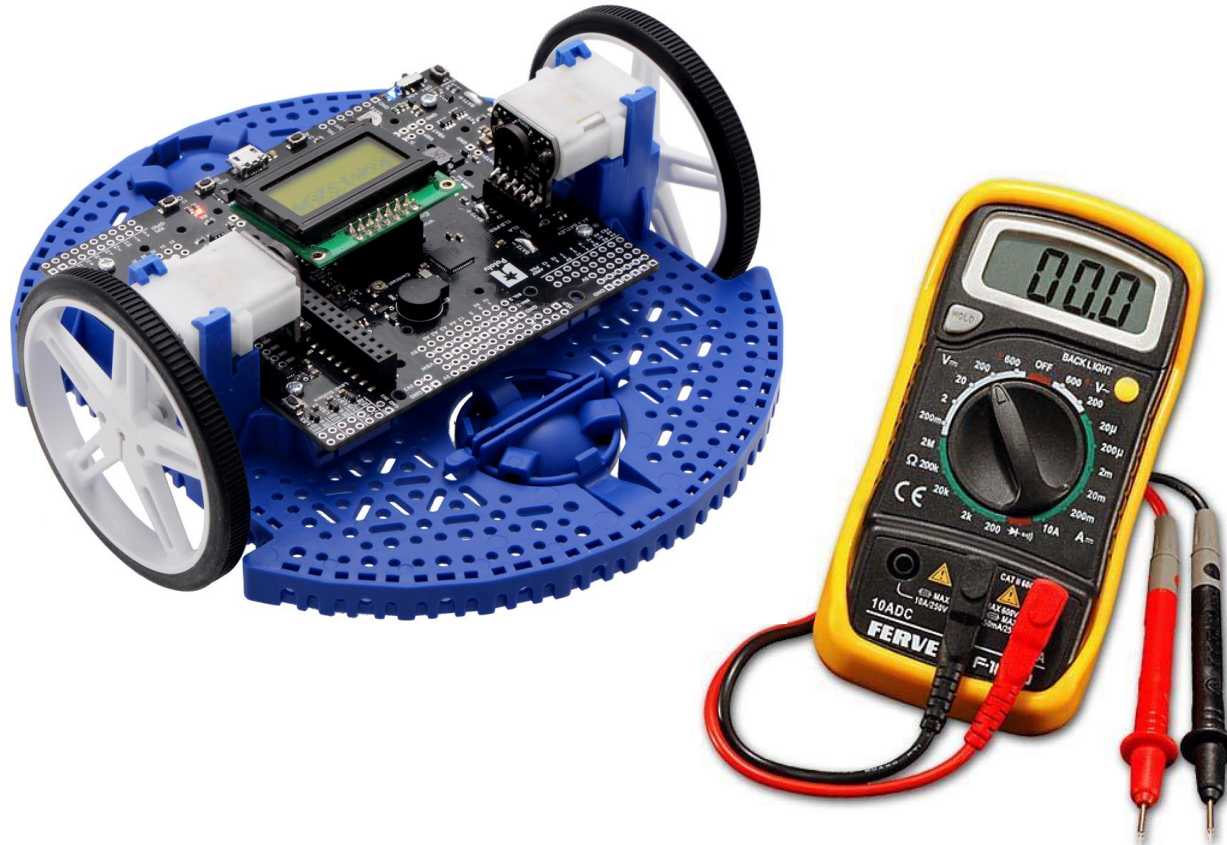
Photoresistor

- Great! a part that changes its resistance based on light!
- **BUT** the romi can't measure resistance directly



Voltage Measurement

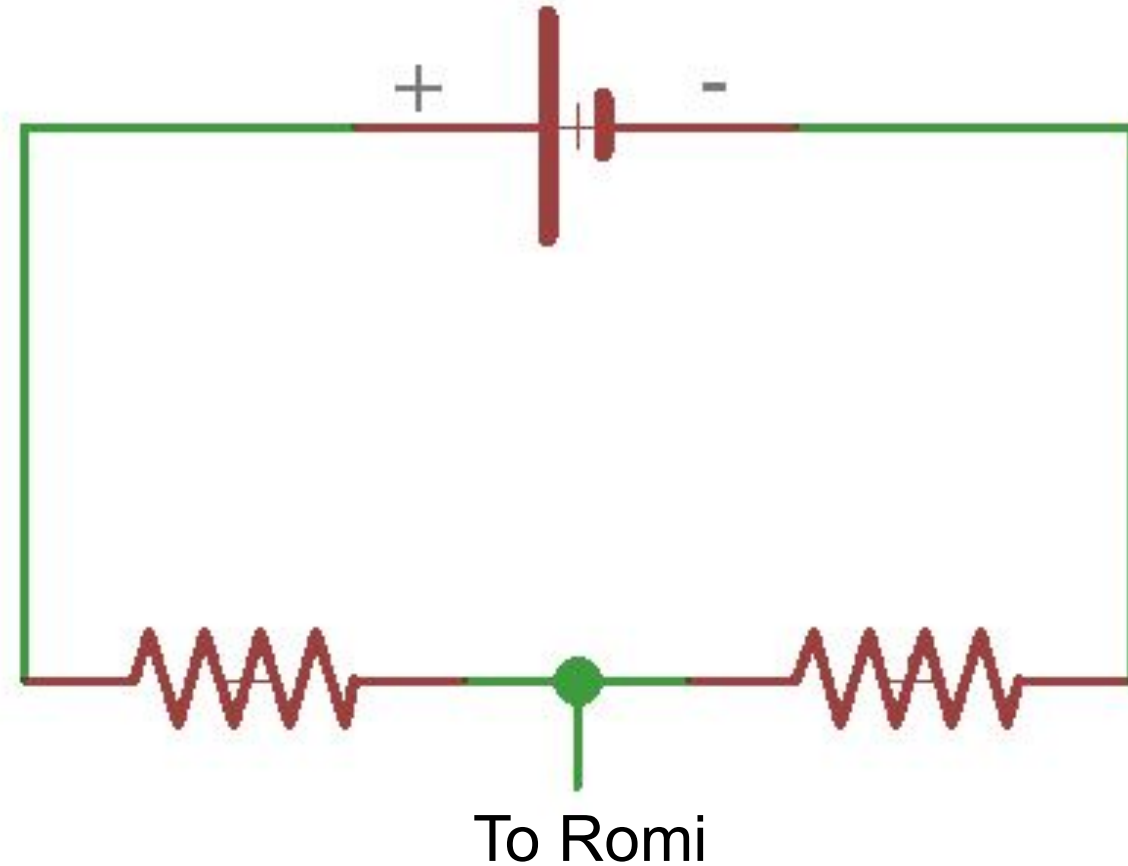
- So what can the Romi Measure?... VOLTAGE!
 - Well, mostly... can only measure voltage from 0V-5V



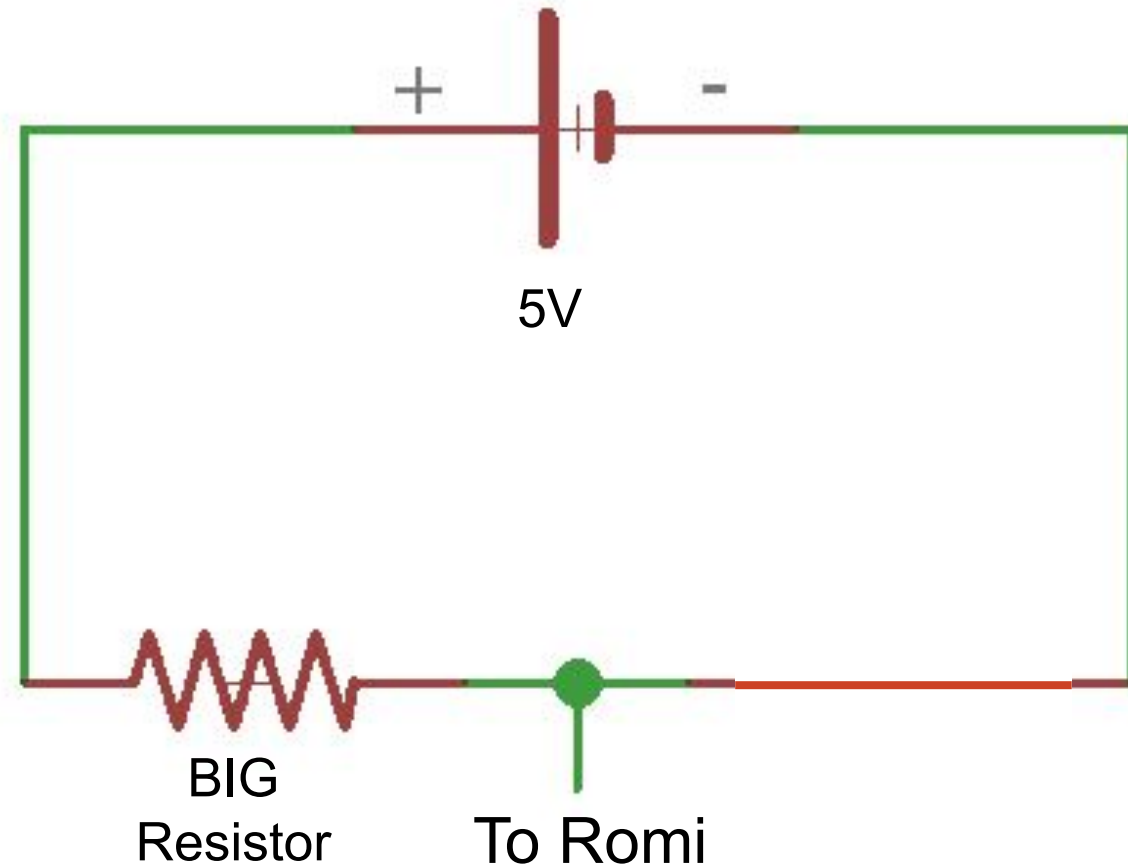
Voltage Divider

- We can use a voltage divider circuit to convert resistance changes into voltage changes that the Romi can read

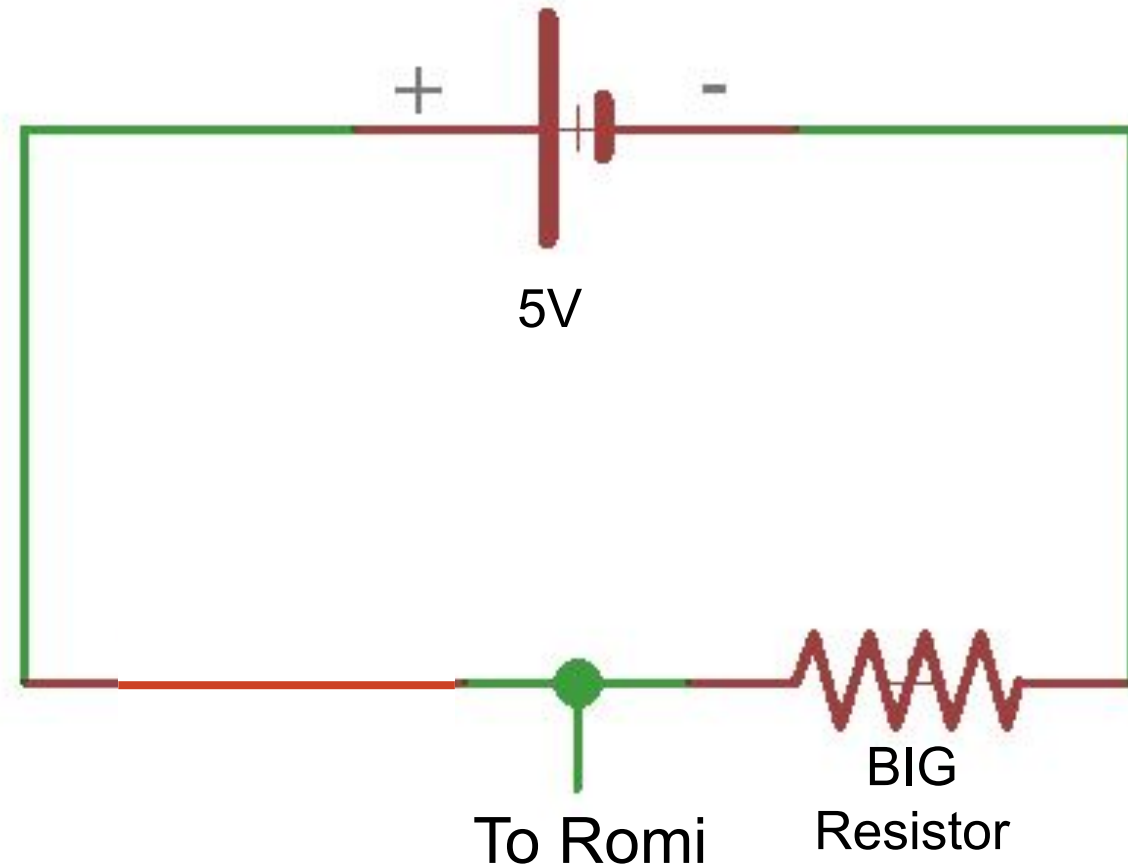
Voltage Divider



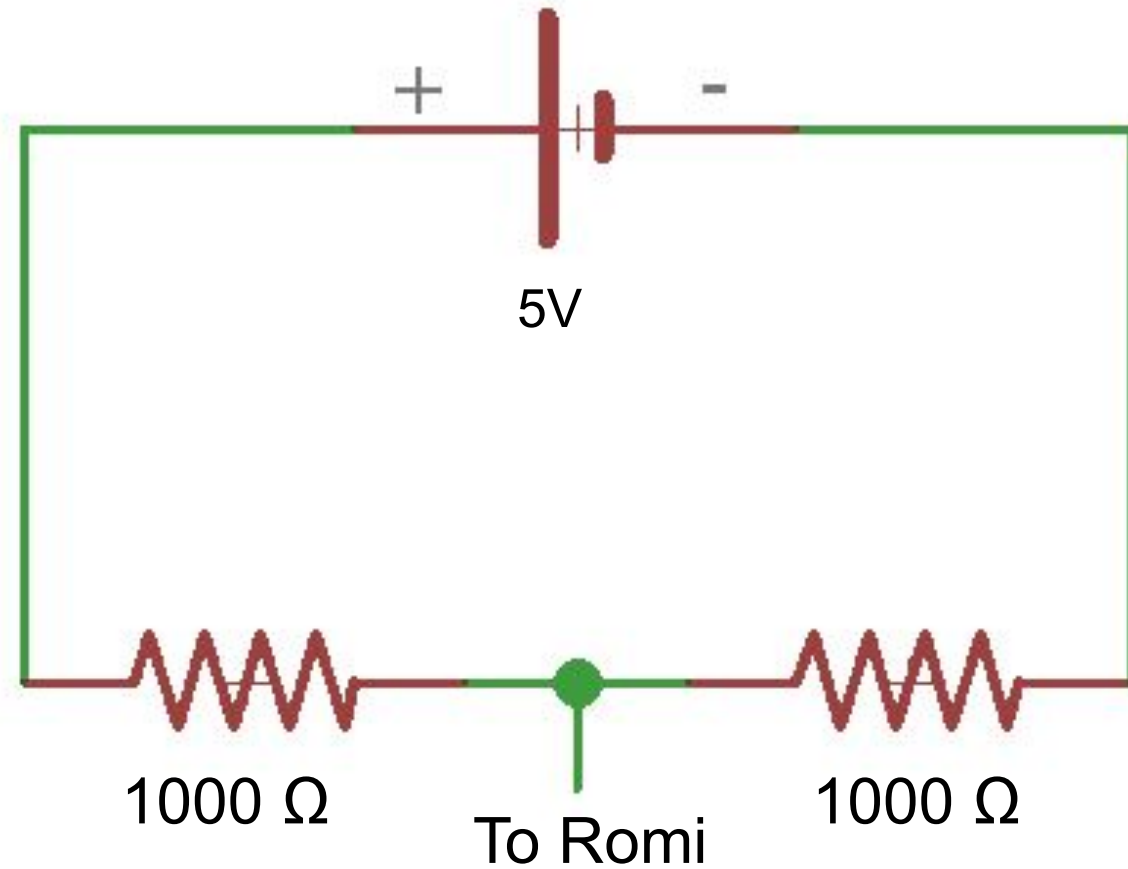
Voltage Divider



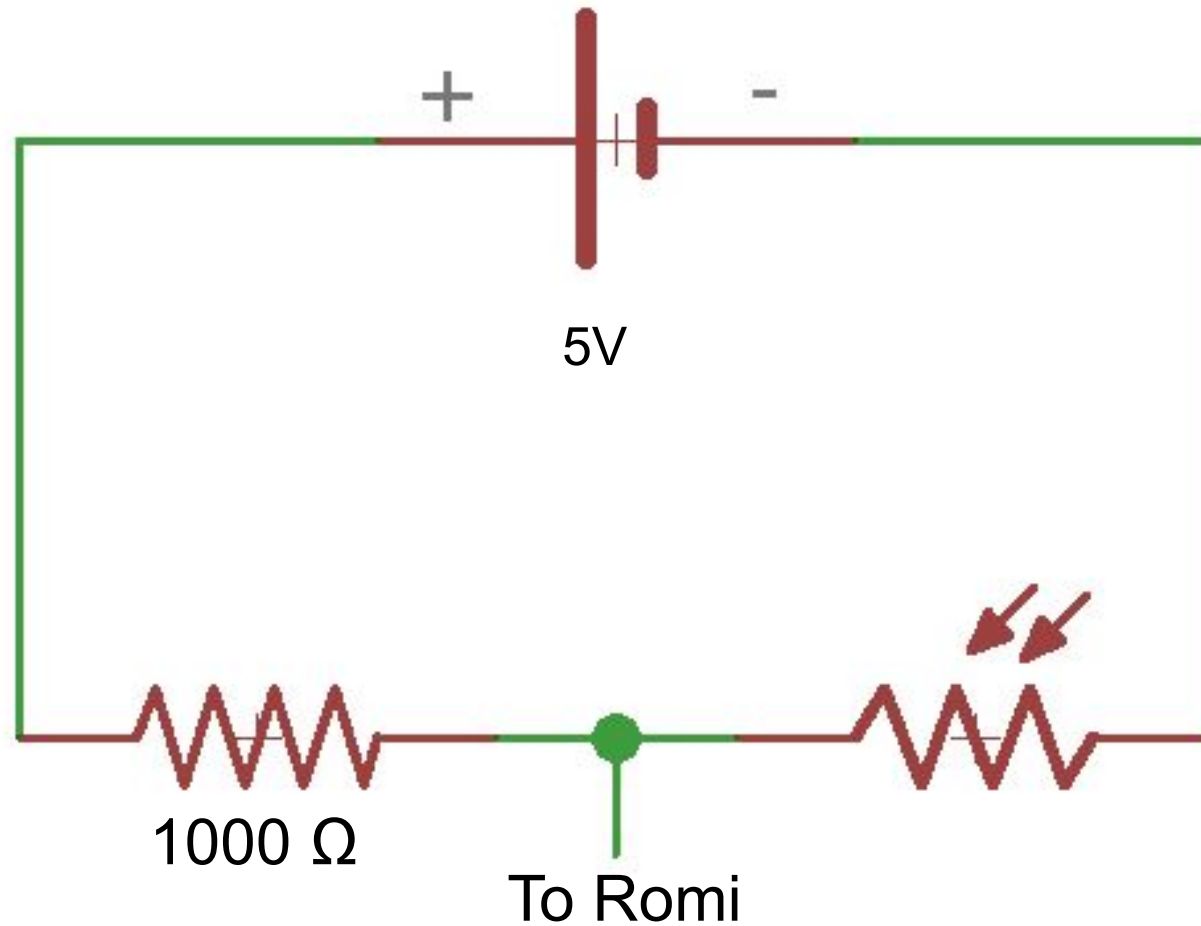
Voltage Divider



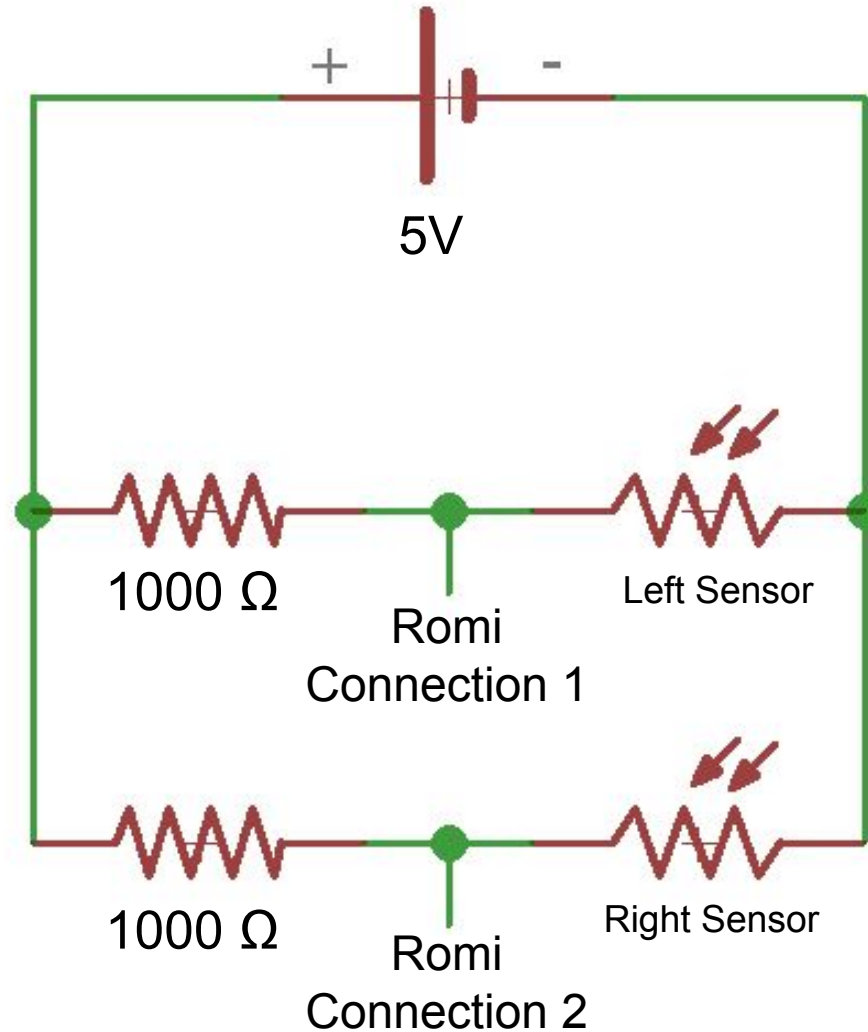
Voltage Divider



Voltage Divider

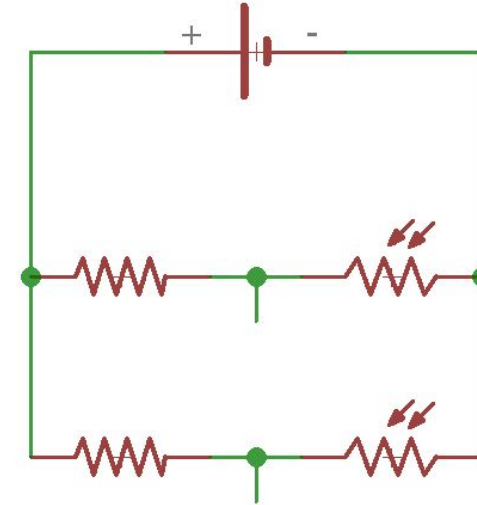


Full Romi Circuit

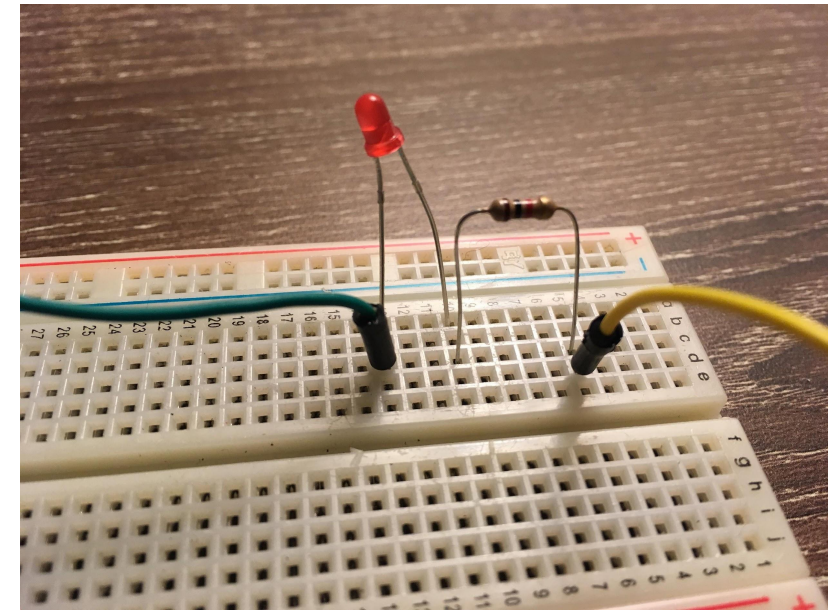
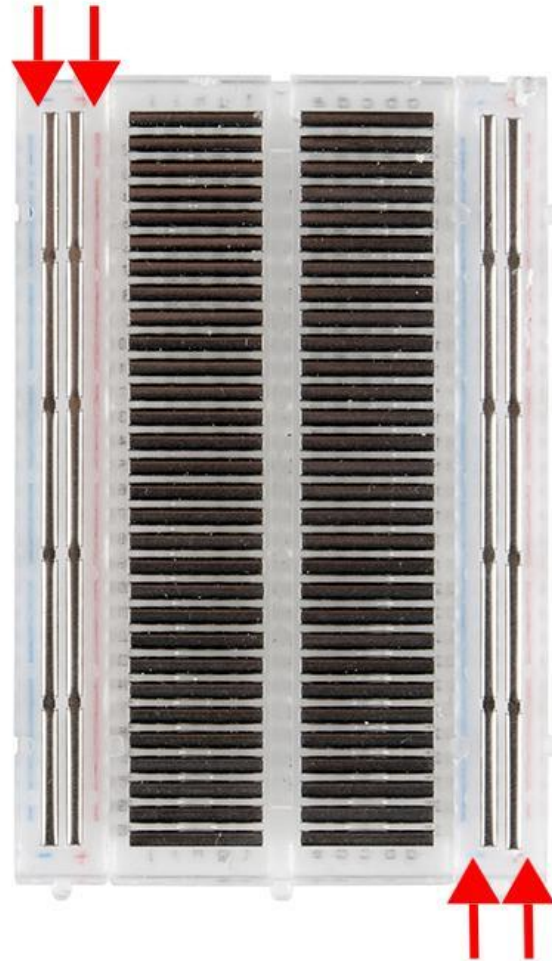
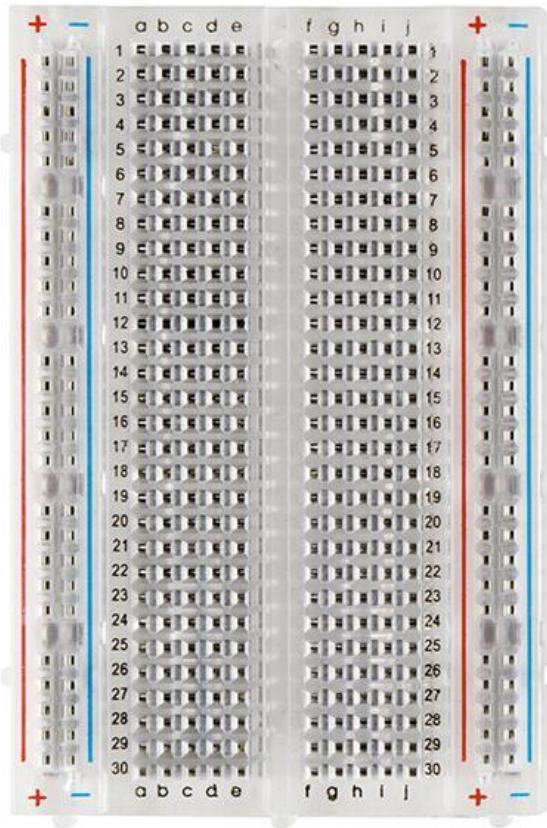


Circuit

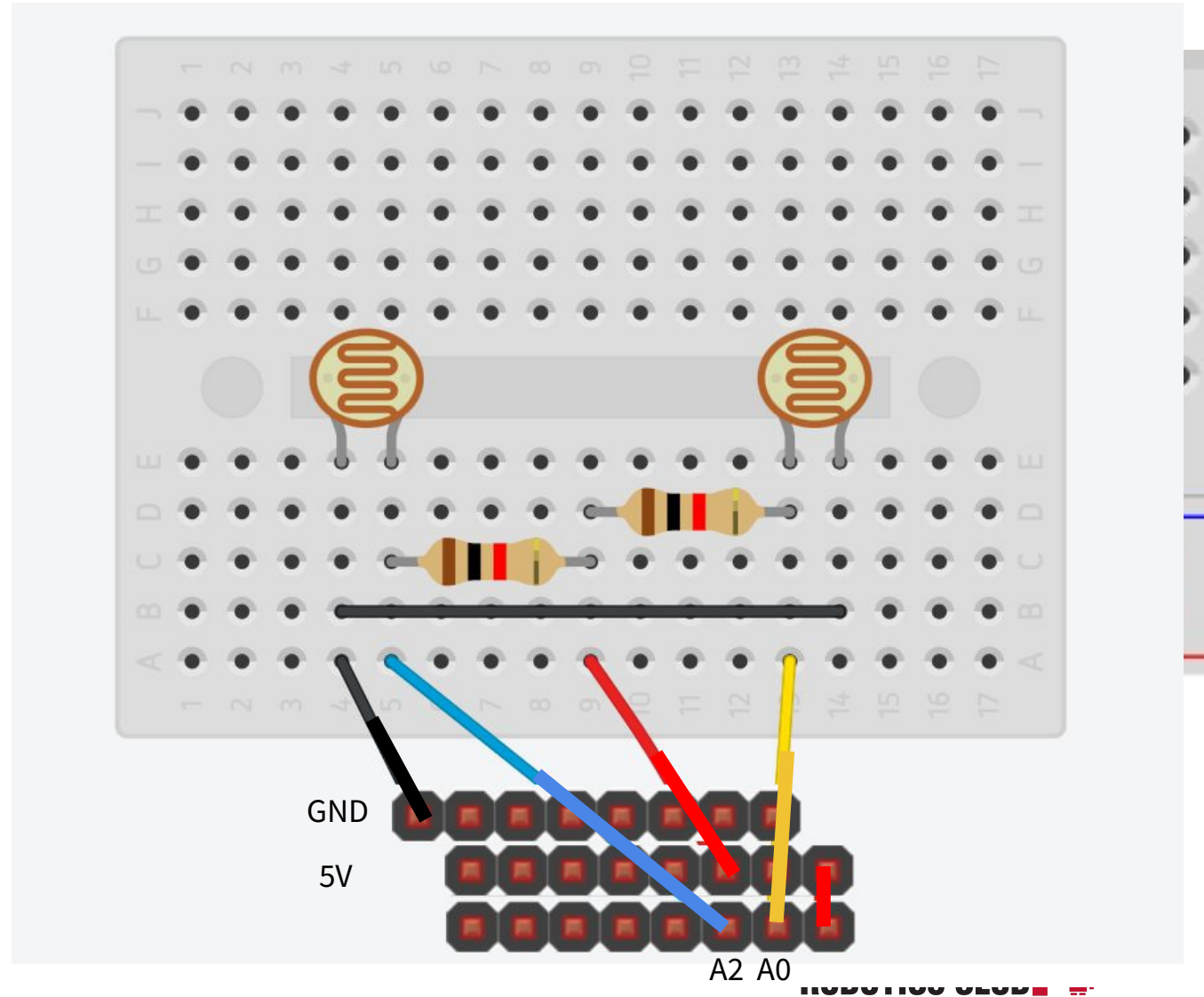
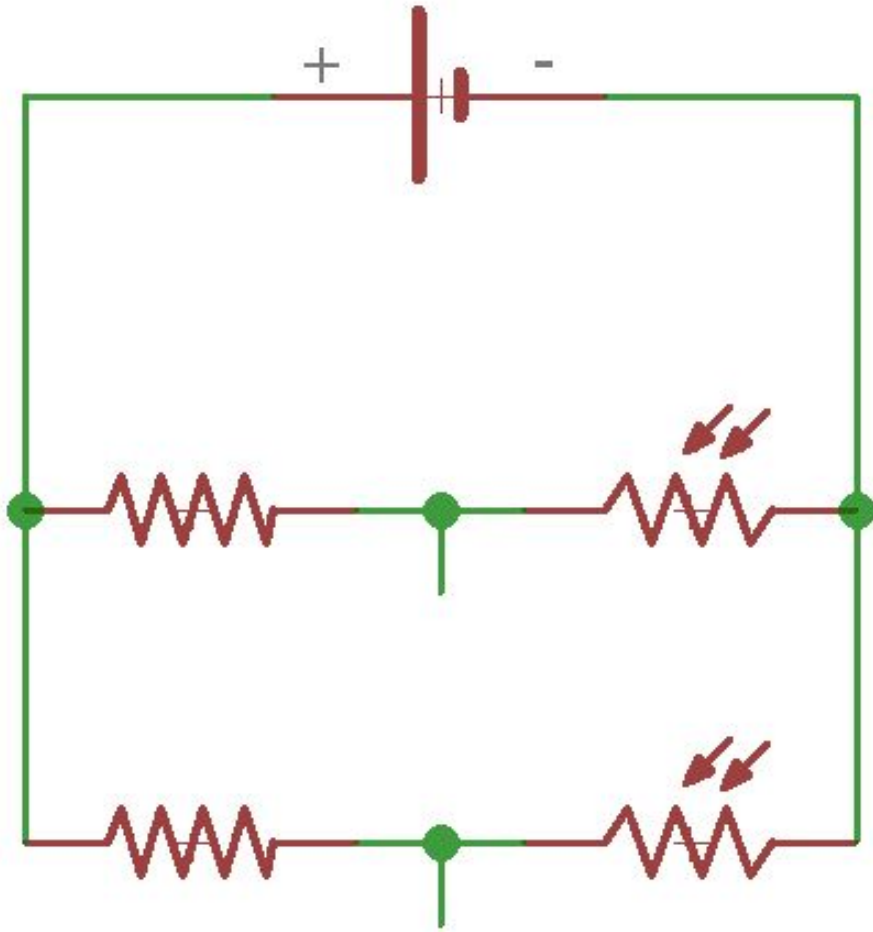
Great! We have our circuit to measure light. But how do we build it?



Breadboards



Build it!



Code

```
#define LEFT_SENSOR A0
#define RIGHT_SENSOR A2

void setup() {
}

void loop() {
    // Sensor readings will be between 0 and 1023
    // 0 = 0V; 1023 = 5V at center of voltage divider
    int left_light = analogRead(LEFT_SENSOR);
    int right_light = analogRead(RIGHT_SENSOR);
}
```